



TS ICET 23rd May 2019 Shift-2

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Analytical Ability

Instructions

In questions numbered 1 to 20, a question is followed by data in the form of two statements labeled as I and II. You must decide whether the data given in the statements are sufficient to answer the questions. Using the data make an appropriate choice from (1) to (4) as per the following guidelines:

Question 1

p and q are positive integers. What is the number of positive divisors of p?

I. $p = q^{10}$

II. q is a Prime number

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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Question 2

If p and q are positive integers whose product is 72. then what is the value of p+q?

I. $p > q$

II. $|p - q| = 1$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

Question 3

If n and m are integers, is $n > m$?

I. $3n + 2m = 11$

II. $n - m = 1$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statement alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

Question 4

Is there positive integer n divisible by 35?

I. 15 divides n.

II. 28 divides n.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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Question 5

What is the value of $\sin \theta + \cos \theta$?

- I. $\sin 2\theta = \frac{1}{2}$
- II. $\operatorname{cosec} \theta + \sec \theta = 5$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

Question 6

In the triangle POR, what is the length of the side OR?

- I. $PQ = 8 \text{ cm}$, $PR = 5 \text{ cm}$
- II. Area of a triangle PQR = 10 Sq.cm.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

Question 7

How much is the loss percent?

- I. The loss is 20% of the selling price.
- II. The cost price is Rs.250

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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Question 8

If n is an integer and 101. The sum of the digits in 7 is 12.
II. One digit in 7 is 3 times the other.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

Question 9

For two sets A and B, what is $n(A \cap B)$?

I. $n(A) = 10$ and $n(B - A) = 9$

II. $n(A - B) = 7$, $n(B - A) = 9$ and $n(A \cup B) = 19$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

Question 10

What percent of 16 is x ?

I. 200 percent of x is less than 1.

II. x is 5 percent of 10.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: B

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Question 11

What is the value of $x^2 - y^2$?

I. $(x - 2018)^2 + (y + 2019)^2 = 0$

II. $(x + 2018)^2 + (y - 2019)^2 = 0$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.

- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

Question 12

What is the value of x ?

I. $3^{48} \equiv x \pmod{10}$

II. $0 \leq x \leq 9$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

Question 13

Will B take more than 9 hours to complete the job alone?

I. A works faster than B.

II. A and B can together finish the job in 6 hours.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

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Question 14

What is the area of the rhombus?

I. Each of its sides is 120 cm.

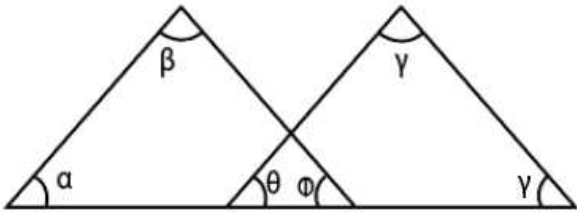
II. Each of its opposite angles are 60° .

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

Question 15

In the figure given below, what is the value of $\alpha + \beta + \gamma + \delta$?



- I. $\alpha + \beta + \gamma + \delta$
- II. $\theta + \phi = 90^\circ$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: D

Question 16

What is the day of the week was 31st December of that year?

- I. In that year the first of March was Monday.
- II. That year was a leap year.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: A

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Question 17

A, B, C, D and E are sitting around a circular table facing the centre. Who is sitting to the immediate left of C?

- I. Only A is between E and B.
- II. D is to the immediate left of B.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

Question 18

What is the area of the sector 4 of the circle?

- I. The sector 4 makes an angle 60° at the centre.
- II. Length of the arc of the sector A is $\frac{110}{21}$ cm.

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

Question 19

What is the area of the right angled triangle?

I. Its longest side is 6 cms.

II. One of its angle is 45° .

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

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Question 20

What is the value of $8a^3 + b^3 - 27c^3$

I. $abc = 10$

II. $2a + b = 3c$

- A if the statement I alone is sufficient to answer the question.
- B if the statement II alone is sufficient to answer the question.
- C if both the statements I and II are sufficient to answer the question but neither statements alone is sufficient.
- D if both the statements I and II together are not sufficient to answer the question and additional data is required.

Answer: C

Instructions

In each of the questions a sequence of numbers or letters that follow a definite pattern is given. Each question has a blank space. This has to be filled by the correct answer from the four given options to complete the sequence without breaking the pattern.

Question 21

100 : 111 :: 144 :

- A 156
- B 149
- C 154
- D 157

Answer: D

Question 22

182: :: 210: 380

- A 156
- B 240
- C 272
- D 342

Answer: D

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Question 23

..... :122::168:290

- A 43
- B 48
- C 50
- D 49

Answer: B

Question 24

12:132::14:

- A 182
- B 172
- C 184
- D 174

Answer: A

Question 25

12 : 1872 :: : 2366

- A 14
- B 18
- C 13
- D 16

Answer: C

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Question 26

corden : zrogbq : : :pxivro

- A mulmul
- B sulsul
- C munmun
- D srspql

Answer: B

Question 27

COUNSEL: BITIRAK: : GUIDANCE :

- A EOHYZKBB
- B FOHYZJBB
- C FPHZZKAB
- D HOHYBJBA

Answer: B

Question 28

$D \times I : 19 \times 9 :: K \times F : \dots\dots\dots$

- A 5×15
- B 15×5
- C 16×21
- D 21×16

Answer: A

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Question 29

3:13::17:

- A 29
- B 23
- C 37
- D 31

Answer: D

Question 30

1331 : 11 :: : 22

- A 3432
- B 4234
- C 6324
- D 10648

Answer: D

Instructions

In the following questions pick the odd thing out:

Question 31

- A Mist
- B Cloud
- C Vapour
- D Rain

Answer: D

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Question 32

- A $\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$
- B $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$
- C $\begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$
- D $\begin{bmatrix} 1 & 4 \\ 4 & 16 \end{bmatrix}$

Answer: A

Question 33

- A 54
- B 53
- C 55
- D 56

Answer: B

Question 34

- A Tiger

- B Deer
C Lion
D Fox



Answer: B

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Question 35

- A $x^2 - 0.2x + 0.01$
B $x^2 + 2x + 1$
C $x^2 + 6x + 9$
D $x^2 - 2x + 2$

Answer: D

Instructions

Each of the questions from 36 to 45 follow a definite pattern. Observe the same and fill in the blanks with suitable answers.

Question 36

6, 10, 18, 30, 46, 66,

- A 94
B 86
C 80
D 90

Answer: D

Question 37

$11\frac{1}{9}, 12\frac{1}{2}, \dots, 16\frac{2}{3}, 20$

- A $13\frac{6}{7}$
B $14\frac{1}{7}$
C $14\frac{2}{7}$
D $13\frac{1}{8}$

Answer: C

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Question 38

E, G, K,, Q, S,

- A M,W
- B M,U
- C L,W
- D N,U

Answer: A

Question 39

768921350, 68921350, 6892135, 892135,

- A 92135
- B 89215
- C 89213
- D 82135

Answer: C

Question 40

11, 30, 128,346,

- A 1221
- B 1331
- C 1224
- D 1334

Answer: D

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Question 41

11,22,35,50,67,

- A 88
- B 84
- C 86
- D 82

Answer: C

Question 42

18,20,17,21,16,22,15,.....

- A 23
- B 24

C 26

D 25

Answer: A

Question 43

$\left\{\frac{1}{5}, \frac{1}{6}, \frac{1}{7}\right\}, \left\{\frac{1}{8}, \frac{1}{11}, \frac{1}{14}\right\}, \left\{\frac{1}{17}, \frac{1}{22}, \frac{1}{27}\right\}, \dots$

A $\left\{\frac{1}{29}, \frac{1}{34}, \frac{1}{39}\right\}$

B $\left\{\frac{1}{28}, \frac{1}{36}, \frac{1}{43}\right\}$

C $\left\{\frac{1}{32}, \frac{1}{38}, \frac{1}{46}\right\}$

D $\left\{\frac{1}{32}, \frac{1}{39}, \frac{1}{46}\right\}$

Answer: D

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Question 44

$(a+b), (a^2-b^2), (a+b)^2(a-b), (a^2-b^2)^2, (a+b)^3(a-b)^2, \dots$

A $(a^2-b^2)^3$

B $(a^2-b^2)^2$

C $(a^2-b)^3$

D $(a-b)^3(a+b)^2$

Answer: A

Question 45

CGK, OSW, AEI

A LQU

B MPU

C MQV

D MQU

Answer: D

Instructions

Study the following table carefully and answer the questions.

The numbers given in the table represent the number of students from various schools playing different games.

Students					
Schools → Games ↓	A	B	C	D	E
Foot ball	125	250	100	175	250
Basket ball	175	200	195	245	225
Cricket	250	200	225	215	200
Tennis	240	210	200	130	165
Badminton	75	125	55	45	100

Question 46

If 20% of the students playing football from school 'A' also play Badminton, then the total number of students playing Badminton from School 'A' is.

- A 120
- B 95
- C 100
- D 110

Answer: C

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Question 47

The number of students playing Basketball from School 'C' is approximately what percent of the number of students playing basketball from School 'E'?

- A 70%
- B 75%
- C 87%
- D 94%

Answer: C

Question 48

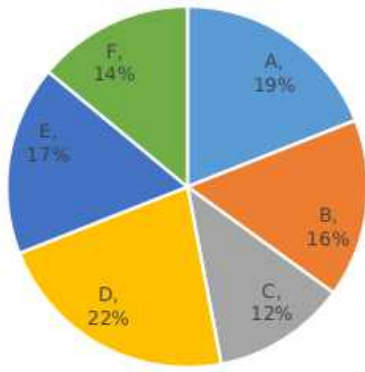
The difference between the average number of students playing all the games from school 'B' and the number of students playing badminton from that school is:

- A 72.
- B 69
- C 65
- D 78

Answer: A

Instructions

The Pie chart given below shows the percentage-wise break up of students studying various specializations A.B.C.D.E and F in M.B.A. course in a university. The total number of students who took admission in MBA in that university is 8000. Study the chart carefully and answer the questions given below.



Question 49

The total number of students studying the specializations in C, D and F is

- A 4080
- B 3840
- C 4000
- D 4240

Answer: B

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Question 50

Students studying B as specialization forms approximately what percent of students studying A as specialization?

- A 84
- B 83
- C 82
- D 85

Answer: A

Question 51

The total number of students studying E as specialization.

- A 1760
- B 1520
- C 1280
- D 1360

Answer: D

Question 52

The ratio of the students studying C as specialization to students studying F as specialization is.

- A 4:7

- B 6:7
C 11:19
D 18:13

Answer: B

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Question 53

If $x : y$ represent the ratio of the number of students studying F and C to the number of students studying D, then $x - y =$

- A $\frac{1}{15}$
B $\frac{2}{17}$
C $\frac{1}{11}$
D $\frac{3}{14}$

Answer: A

Instructions

Study the following data and answer the questions.

"In a study of 1500 rivers the following data were reported: 520 rivers were polluted by sulphur compounds, 335 were polluted by phosphates, 425 were polluted by crude oil, 100 were polluted by both crude oil and sulphur compounds; 180 were polluted by both sulphur compounds and phosphates; 150 were polluted by both phosphates and crude oil and 28 were polluted by sulphur compounds, phosphates and crude oil."

Question 54

The number of the rivers polluted by at least one of the three impurities is

- A 878
B 359
C 781
D 504

Answer: A

Question 55

The number of the rivers polluted by exactly one of the three impurities is

- A 504
B 878
C 492
D 638

Answer: A

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Instructions

In a certain code, the r^{th} letter of the English alphabet is coded as $r + 10 (\equiv \text{mod } 26)^{th}$ letter cyclically. For example, 4th letter D is coded as 14th letter N and 24th letter X is coded as 8th letter H (because $r + 10 (\equiv \text{mod } 26)$). The reverse process is used for decoding. Basing on this coding and decoding, answer questions.

Question 56

'ARMY' is coded as

- A KBWH
- B KBWJ
- C KBVI
- D KBWI

Answer: D

Question 57

'SEVEN' is coded as

- A COFOY
- B COGOY
- C COFOX
- D COEOX

Answer: C

Question 58

Which word is coded as 'MYVN'*?

- A COIN
- B BOLD
- C COLD
- D COME

Answer: C

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Question 59

The code for 'TSICET' is

- A ECSMPE
- B DCTMOD
- C DCSMPD

D DCSMOD

Answer: D

Question 60

Which word is coded as 'GKB'?

A WAS

B WAR

C CAT

D YET

Answer: B

Instructions

In a certain code language, '481' means "sky is blue", '246' means "sea is deep" and '698' means "sea looks blue". Using this information answer the questions.

Question 61

What is the code for "deep"?

A 1

B 2

C 4

D 6

Answer: B

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Question 62

The code for 'sea sky looks' is

A 619

B 492

C 846

D 298

Answer: A

Question 63

The code for 'Sky blue is deep' is:

A 6281

B 8194

C 1824

D 1842

Answer: D

Question 64

692 is the code for

A deep is sea

B sky is blue

C blue looks deep

D sea looks deep

Answer: D

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Question 65

What is the code for "blue sea" ?

A 19

B 68

C 86

D 91

Answer: C

Instructions

For the following questions answer them individually

Question 66

What day of the week was 26th January 1983?

A Monday

B Wednesday

C Thursday

D Friday

Answer: B

Question 67

If a clock takes 22 seconds to strike 12. how many seconds will it take to strike 9?

A $16\frac{1}{2}$ sec

B 18 sec

C 16 sec

D 17 sec

Answer: C

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Question 68

At what time between 2 O'clock and 3 O'clock will the two hands in a clock be at right angle to each other?

- A $2.25\frac{5}{11}$
- B $2.23\frac{3}{11}$
- C $2.26\frac{3}{11}$
- D $2.27\frac{3}{11}$

Answer: D

Question 69

Introducing a man, a woman said "he is the only son of my mother's mother", How is the woman related to the man?

- A Niece
- B Sister
- C Aunt
- D Mother

Answer: A

Question 70

Reaching the place of meeting 15 minutes before 8.30 a.m. Ram found himself half an hour earlier than the man who was 40 minutes late to the meeting. Then the scheduled time of the meeting is.

- A 8:20 a.m.
- B 8:15 a.m.
- C 8:10 a.m.
- D 8:05 a.m.

Answer: D

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Question 71

If t_1 is the time elapsed between 9:25 a.m to 2:50 p.m, and if t_2 is the time elapsed between 10:10 a.m to 5:15 p.m then $t_1 : t_2 =$

- A 17 : 13
- B 13 : 17
- C 53 : 73
- D 73 : 53

Answer: B

Question 72

Six persons P, O, R, S, T and U are sitting around a roundtable facing the centre of the table. T is between P and S, R is to the immediate left of U and to the immediate right of O and O is to the immediate right of S. Who is sitting between R and S?

- A T
- B S
- C R
- D Q

Answer: D

Question 73

If α denotes multiplication, β denotes division, γ denotes addition and δ denotes subtraction, then,

$$6\delta\{[(6\gamma 4)\beta 8]\alpha 2\} =$$

- A $2\frac{1}{2}$
- B $3\frac{1}{2}$
- C 3
- D 2

Answer: B

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Question 74

For any real numbers a and b, if $a*b = a + b - ab$ then $(1*2)*(3*4) =$

- A 1
- B 2
- C 3
- D 4

Answer: A

Question 75

If Z denote the set of integers then the number of elements in the set $A = \{(a, b) : 2a^2 + 3b^2 = 25; a, b \in Z\}$ is

- A 8
- B 4
- C 2
- D 12

Answer: A

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Mathematical Ability

Instructions

For the following questions answer them individually

Question 76

$$x^y = y^x \Rightarrow \left(\frac{x}{y}\right)^{\frac{y}{x}} =$$

A 0

B 1

C $x^{\frac{y}{x}-1}$

D $x^{1-\frac{y}{x}}$

Answer: D

Explanation:

given that

$$x^y = y^x$$

so we can write that $y = x^{\frac{y}{x}}$

and substituting the value to

$$x^{\frac{y}{x}}$$

we get $\frac{x}{x^{\frac{y}{x}}} = \frac{x^{\frac{y}{x}}}{x^{\frac{y}{x}}}$

$$= x^{\frac{x}{y} - \frac{x}{y} \cdot \frac{y}{x}} = x^1$$

$$x^{\frac{x}{y} - 1} \text{ answer}$$

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Question 77

$$x^{x\sqrt{x}} = (x\sqrt{x})^x \Rightarrow x = \dots\dots\dots$$

A $\frac{3}{2}$

B $\frac{9}{4}$

C 1

D $\frac{1}{2}$

Answer: B

Explanation:

$$x^{x\sqrt{x}} = (x\sqrt{x})^x \Rightarrow$$

Question 78

$$\frac{1}{a} : \frac{1}{b} : \frac{1}{c} = 3 : 5 : 7 \Rightarrow a : b : c =$$

- A 30 : 15 : 21
- B 32 : 18 : 12
- C 35 : 21 : 15
- D 37 : 23 : 17

Answer: C

Explanation:

$$\frac{1}{a} : \frac{1}{b} : \frac{1}{c} = 3 : 5 : 7 \Rightarrow a : b : c = ?$$

now we have from above eq $a : \frac{1}{3}, b : \frac{1}{5}, c : \frac{1}{7}$

let take LCM of 3,5 and 7 we get lcm=105

$$\text{now we have } a : b : c := \frac{1}{3} \times 105 : \frac{1}{5} \times 105 : \frac{1}{7} \times 105$$

after solving we get

$$a : b : c := 35 : 21 : 15 \text{ answer}$$

Question 79

If for the natural numbers $a, b, c, a^3 + b^3 + c^3 = 8072$ and $a : b = b : c = 3 : 2$, then $a =$

- A 12
- B 14
- C 16
- D 18

Answer: D

Explanation:

$$a, b, c, a^3 + b^3 + c^3 = 8072 \text{ and } a : b = b : c = 3 : 2 \text{ then } a = ?$$

$$\text{let } \frac{a}{b} = \frac{3}{2} \text{ times } \frac{3x}{3} \text{ and } \frac{b}{c} = \frac{3}{2} \text{ times } \frac{2x}{2}$$

then we get

$$a = 9x, b = 6x, c = 4x$$

now put the value of a,b, and c in equation $a, b, c, a^3 + b^3 + c^3 = 8072$

$$\text{now we have eq } 9x^3 + 6x^3 + 4x^3 = 8072$$

$$\text{then we get } 729x^3 + 216x^3 + 64x^3 = 8072$$

$$\text{then } 1009x^3 = 8072$$

$$x = 2$$

put the value of x and we get

$$a = 18, b = 12, c = 8 \text{ answer}$$

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Question 80

$$\frac{1}{\sqrt{5}-1} - \frac{1}{2-\sqrt{3}} + \frac{1}{\sqrt{3}-\sqrt{2}} - \frac{1}{\sqrt{2}-1} + 1 =$$

- A 0
- B 1
- C $\sqrt{5}$
- D $2\sqrt{5}$

Answer: C

Question 81

$$\sqrt{18 + x\sqrt{2}} = \sqrt{12} + \sqrt{6} \Rightarrow x =$$

- A 6
- B 8
- C 12
- D 18

Answer: C

Explanation:

$$\sqrt{18 + x\sqrt{2}} = \sqrt{12} + \sqrt{6} \Rightarrow x =$$

Take square both side

$$(\sqrt{18 + x\sqrt{2}})^2 = (\sqrt{12} + \sqrt{6})^2 \Rightarrow x =$$

$$18 + x\sqrt{2} = 12 + 6 + 2 * \sqrt{12} * \sqrt{6} \Rightarrow x =$$

$$18 + x\sqrt{2} = 18 + 2 * \sqrt{6} * \sqrt{2} * \sqrt{6} \Rightarrow x =$$

18 will be eliminated from both the sides.

$$x\sqrt{2} = 2 * \sqrt{6} * \sqrt{2} * \sqrt{6}$$

$$x = 2 * 6$$

$$x = 12$$

Hence, option C is correct.

Question 82

When a positive integer is divided by 783 the remainder was 48. What will be the remainder when the same number is divided by 29 ?

- A 13
- B 17
- C 19
- D 23

Answer: C

Explanation:

Let the number be N.

As the number leaves 48 as remainder when divided by 783, then

$$N = 783 * k + 48 \text{ (Here, k is an integer)}$$

$$N = 29 * 27 * k + 29 + 19$$

$$N = 29(27k + 1) + 19$$

$$\text{Let } 27k + 1 = m$$

$$\text{then } N = 29m + 19$$

So, remainder = 19

Hence, option C is correct.

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Question 83

The numbers of divisors of 1248 other than 1 and the number itself is:

A 16

B 18

C 22

D 24

Answer: C

Explanation:

Factors of 1248 = $2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 13$

It can be written as

$$1248 = 2^5 \times 3^1 \times 13^1$$

Factors are written in the form: $2^p \times 3^q \times 13^r$

Total number of divisors = $(p+1) \times (q+1) \times (r+1)$

$$= (5+1) \times (1+1) \times (1+1)$$

$$= 6 \times 2 \times 2 = 24$$

1 and the number itself are excluded, then

$$24 - 2 = 22$$

So, the number of divisors will be 22.

Hence, option C is correct.

Question 84

What is the least number which when subtracted from 3000 is exactly divisible by 7, 11, 13?

A 898

B 996

C 997

D 998

Answer: D

Explanation:

Take L.C.M. of 7, 11, 13

$$\text{L.C.M. (7, 11, 13)} = 1001$$

$$3000 = 1001 \times m + k \text{ (where } k \text{ is remainder)}$$

Put value of $m = 2$

$$3000 = 1001 \times 2 + k$$

$$3000 = 2002 + k$$

$$k = 998$$

So, the least number which when subtracted from 3000 is exactly divisible by 7, 11, 13.

Hence, option D is correct.

Question 85

What is the largest number which will divide 1354, 1866 and 2762 leaving 10 as remainder in each case?

A 32

B 48

C 64

D 128

Answer: C

Explanation:

Since 10 is remainder in each case, subtract 10 from every number.

$$1354 - 10 = 1344$$

$$1866 - 10 = 1856$$

$$2762 - 10 = 2752$$

$$\text{HCF}(1344, 1856, 2752) = 64$$

So, 64 will be the greatest number.

Hence option C is correct.

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Question 86

If the recurring decimal number $3.\overline{784}$ is equal to the rational number $\frac{p}{q}$ and $\text{GCD}(p, q) = 1$, then $p + q =$

A 1057

B 1579

C 3118

D 4737

Answer: B

Explanation:

$$\text{Let } x = 3.\overline{784}$$

We can write it as

$$x = 3.7848484\ldots$$

Multiply the equation with 10 on both the sides

$$10x = 37.8484\ldots \text{ equation 1}$$

Multiply equ 1 with 100 on both the sides

$$1000x = 3784.8484\ldots \text{ equation 2}$$

Subtract equation 1 from equation 2

$$1000x - 10x = 3784.8484... - 37.8484...$$

$$990x = 3747$$

$$x = \frac{3747}{990}$$

where $\frac{3747}{990}$ is in $\frac{p}{q}$ form

$$\text{For } GCD(p, q) = 1$$

$$\frac{3747}{990} = \frac{1249}{330}$$

$$p + q = 1249 + 330$$

$$= 1579$$

Hence, option B is correct answer.

Question 87

$$3 + \frac{1\frac{1}{5} + \frac{1}{6}}$$

A $\frac{421}{130}$

B $\frac{431}{130}$

C $\frac{441}{150}$

D $\frac{451}{160}$

Answer: A

Explanation:

$$3 + \frac{1\frac{1}{5} + \frac{1}{6}}$$

$$3 + \frac{1\frac{1}{5} + \frac{1}{6}}$$

$$3 + \frac{1}{4} + \frac{1}{31}$$

$$3 + \frac{12\frac{1}{4} + 6}{31}$$

$$3 + \frac{31}{130}$$

$$\frac{421}{130}$$

Hence, option A is correct.

Question 88

The largest among the following numberts:

A $\sqrt{2}$

B $\sqrt[3]{3}$

C $\sqrt[5]{5}$

D $\sqrt[7]{7}$

Answer: A

Explanation:

$$\sqrt{2}, \sqrt[3]{3}, \sqrt[5]{5}, \sqrt[7]{7}$$

$$2^{\frac{1}{2}}, 3^{\frac{1}{3}}, 5^{\frac{1}{5}}, 7^{\frac{1}{7}}$$

Now, take L.C.M. of denominators of power

L.C.M. of (2, 3, 5, 7) = 210

$$2^{\frac{1 \times 105}{2 \times 105}}, 3^{\frac{1 \times 70}{3 \times 70}}, 5^{\frac{1 \times 42}{5 \times 42}}, 7^{\frac{1 \times 30}{7 \times 30}}$$

$$2^{\frac{105}{210}}, 3^{\frac{70}{210}}, 5^{\frac{42}{210}}, 7^{\frac{30}{210}}$$

$$\sqrt[210]{2^{105}}, \sqrt[210]{3^{70}}, \sqrt[210]{5^{42}}, \sqrt[210]{7^{30}}$$

The largest power is of 2. So, $\sqrt[210]{2^{105}}$ will be the largest number.

Hence, option A is correct.

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Question 89

Which of the following statements is true ?

A $\frac{1}{4} > \frac{5}{16} > \frac{7}{18} > \frac{8}{19}$

B $\frac{1}{4} > \frac{7}{18} > \frac{8}{19} > \frac{5}{16}$

C $\frac{8}{19} > \frac{7}{18} > \frac{5}{16} > \frac{1}{4}$

D $\frac{8}{19} > \frac{5}{16} > \frac{1}{4} > \frac{7}{18}$

Answer: C

Explanation:

As, the denominators are 4, 6, 8, 9 in each option

take L.C.M. of 4, 6, 8, 9 = 72

Now, make all the denominators equal to 72.

Since 1 is common in each question, we can ignore it,

Option A: $\frac{1 \times 18}{4 \times 18} > \frac{5 \times 12}{16 \times 12} > \frac{7 \times 9}{18 \times 9} > \frac{8 \times 8}{19 \times 8}$

$$\frac{18}{172} > \frac{60}{172} > \frac{63}{172} > \frac{64}{172}$$

(condition does not satisfy)

Option B: $\frac{1 \times 18}{4 \times 18} > \frac{7 \times 9}{18 \times 9} > \frac{8 \times 8}{19 \times 8} > \frac{5 \times 12}{16 \times 12}$

$$\frac{18}{172} > \frac{63}{172} > \frac{64}{172} > \frac{60}{172}$$

(condition does not satisfy)

Option C: $\frac{8 \times 8}{19 \times 8} > \frac{7 \times 9}{18 \times 9} > \frac{5 \times 12}{16 \times 12} > \frac{1 \times 18}{4 \times 18}$

$$\frac{64}{172} > \frac{63}{172} > \frac{60}{172} > \frac{18}{172}$$

(condition satisfies)

Option D: $\frac{8 \times 8}{19 \times 8} > \frac{5 \times 12}{16 \times 12} > \frac{1 \times 18}{4 \times 18} > \frac{7 \times 9}{18 \times 9}$

$$\frac{64}{172} > \frac{60}{172} > \frac{4}{172} > \frac{63}{172}$$

(condition does not satisfy)

Hence, option C is correct.

Question 90

40 liters of milk contains 10% of water. How many liters of water are to be added to increase the water content to 20%?

- A 3
- B 4
- C 5
- D 6

Answer: C

Explanation:

Given: Amount of mixture = 40L

Water content in the mixture = 10% of 40 = $40 \times \frac{10}{100} = 4\text{L}$

So, milk content in the mixture will be = $40 - 4 = 36\text{L}$

Let the amount of water need to be added to make the water content to 20% be xL.

Then, $\frac{4+x}{40+x} \times 100 = 20$

$$x = 5\text{L}$$

Hence, option C is correct.

Question 91

The salary of an employee was increased by 20% and then reduced by 10%. What was the net change in the employee's salary?

- A 10%increase
- B 8% increase
- C 8%decrease
- D 5% increase

Answer: B

Explanation:

Solution

Let original salary be Rs.100

After increment of 20% on original salary (Rs.100)

Salary = $(100+20)/100 \times \text{Rs.}100 = 1.2 \times \text{Rs. } 100 = \text{Rs. } 120.$

After decrement of 10% on increased salary (Rs.120)

Salary = $(120-10)/100 \times \text{Rs.}120 = 0.9 \times \text{Rs. } 120 = \text{Rs. } 108.$

So the final salary stands at Rs. 108

Net change w.r.t. original salary = $\text{Rs. } (108-100) = \text{Rs. } 8$

% change = $(\text{Rs.}8/100) \times \text{Rs.}100 = 8\%$ **Answer**

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Question 92

If an article is sold at a loss of 9% instead of loss of 15% a dealer gets Rs. 420 more. The cost price of the article (in Rs.) is

- A 6,000

- B 7,000
- C 8,000
- D 9,000

Answer: B

Explanation:

Solution:

Old loss % = 15

New loss % = 9

Difference in loss percentages = 15-9 = 6%

This **6%(of CP)loss** corresponds to monetary loss of **Rs 420**.

1% of CP corresponds to Rs. $420/6 = \text{Rs. } 70$

100% of CP = Rs. $70 \times 100 = \text{Rs. } 7000$ **Answer**

Question 93

A person buys a car at 20% discount of the original price and sells it at 20% higher than the original price. The percentage gain in this transaction is

- A 20
- B 40
- C 50
- D $66\frac{2}{3}$

Answer: C

Explanation:

Solution

Let Original price of the car be Rs. 100

After 20% discount car's price = $(100-20)/100 \times \text{Rs. } 100 = 0.8 \times \text{Rs. } 100$
= Rs. 80.

So CP for Man = Rs. 80

Car was sold at 20% higher price,

So SP = $(100+20)/100 \times \text{Rs. } 100$
= $1.2 \times 100 = \text{Rs. } 120$

SP for man = Rs. 120

Profit % = $(\text{SP}-\text{CP})/\text{CP}$

= $(120 - 80) / 80$

= **50 % Answer**

Question 94

In a business P, O and R invested ₹7,200, ₹12,600 and ₹9,600 respectively. What is the share of R (in ₹) in a total profit of 6,860 ?

- A 2160
- B 2240

C 2280

D 2340

Answer: B

Explanation:

Solution:

The ratio of investments of P, Q and R is 7200:12600:9600.

The standard ratio will be 12:21:16.

So for every profit of Rs. (12+21+16), R will get Rs. 16.

i.e. So for every profit of Rs. 49, R will get Rs. 16.

for Profit of Re 1, R will get Rs. (16/49)

For Profit of Rs 6860, R will get Rs. (16 / 49) x 6860

= **Rs. 2240. Answer**

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Question 95

P and Q enter into a partnership with capitals in the ratio of 3:4. At the end of 10 months P withdraws. If they share profit in the ratio 5:6 for how many months Q's capital was used ?

A 12

B 10

C 9

D 8

Answer: C

Explanation:

Solution

Let P's and Q's initial investment be 3x and 4x respectively.

P's investment was for 10 months so it will be proportional to $(3x)(10) = 30x$

let Q's investment be for 'm' months, so its proportional investment will be $(4x)(m)$.

So according to investment profits will be divided in the ratio $30x : 4mx = 30:4m$

A/q

=> $30:4m = 5:6$

=> $30/4 =$

=> **m = 9**

Hence Q invested for 9 months.

Question 96

A pipe can fill a tank in 15 hours. Another pipe can empty the full tank in 20 hours. If both the pipes are opened together the number of hours required to fill the tank is

A 50

B 54

C 60

D 64

Answer: C

Explanation:

Solution:

Let total capacity of the tank be 60 units [LCM of 15,20]

First pipe fills 60 units (full tank) in 15 hrs. Hence in 1 hr. it fills $(60/15) = 4$ units

Second pipe empties 60 units (full tank) in 20 hrs. Hence in 1 hr. it empties $(60/20)$ units = 3 units or fills $(-)$ 3 units.

If both the pipes are opened simultaneously, total tank filled in 1 hr = $4 + (-)3$ units = 1 unit.

So to fill full capacity (60 units), time taken = $(60/1) = 60$ hrs.

Question 97

Two pipes P and O can fill a tank in 12 minutes and 15 minutes respectively. If both are opened together and P is closed after 3 minutes then the time required to fill the tank (in minutes) is

A $11\frac{1}{4}$

B $11\frac{1}{2}$

C $12\frac{1}{4}$

D $12\frac{1}{2}$

Answer: A

Explanation:

Solution:

Let total work be 60 units [LCM of 12,15]

P does 60 units in 12 minutes. Hence in 1 minute P does $(60/12)$ units = 5 units.

O does 60 units in 15 minutes. Hence in 1 minute O does $(60/15)$ units = 4 units.

Total work in 1 minute = 9 units

Now, for the given scenario,

First,

P and O work for 3 minutes together @ 9 units/min

Work done = (9×3) units = 27

Work remaining = Total units of work - work done = $60 - 27 = 33$ UNITS .

Then,

O work alone @ 4 units/min.

Time taken = $(\text{Work remaining} / \text{speed}) = 33/4$ mins. = 8.25 minutes

Total time = $3 + 8.25 = 11.25$ minutes or $11\frac{1}{4}$ minutes

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Question 98

A Person runs 200 meters in 20 seconds. What is his speed in kmph?

A 28

B 36

C 38

D 40

Answer: B

Explanation:

Solution

Method 1

Distance = 200 m

Time = 20s

Speed in m/s = $(200/20)$ m/s

$$= 10\text{m/s}$$

Now Important pointer to remember

5m/s = 18 km/h

So **10m/s = 36kmph Answer**

Method 2

20 s for 200m.

1 Hr. (20 x 3 x 60 s) for (200 x 3 x 60) m

1 Hr for 36000m

1 Hr for 36 km

Hence speed **36kmph Answer**

Question 99

Running at $\frac{5}{4}$ of his usual speed a person improves his timing by 30 minutes. What was his usual time in hours?

A 3

B $\frac{3}{24}$

C $\frac{1}{22}$

D 2

Answer: C

Explanation:

Solution:

Let old speed and time be S_1 and T_1 .

Let new speed and time be S_2 and T_2 .

A/q,

$$S_2 = \left(\frac{5}{4}\right)S_1 = 1.25 S_1 \text{Eqn (i)}$$

$$T_1 - T_2 = 30\text{minutes} = 0.5 \text{ hr.....Eqn (ii)}$$

As Distance remains constant ; **$S \times T = \text{constant}$** ;

$$S_1 \times T_1 = S_2 \times T_2$$

$$S_1 \times T_1 = (1.25 S_1) \times T_2 \text{.....from Eqn.(i)}$$

$$T_1/T_2 = 1.25$$

$$T_1 = 1.25 T_2$$

and $T_1 - T_2 = 0.5 \text{ hr}$ from Eqn.(ii)

$$1.25T_2 - T_2 = 0.5 \text{ Hr}$$

$$0.25 T_2 = 0.5 \text{ Hr.}$$

$$T_2 = 2 \text{ Hr.}$$

$$T_1 = \text{old time} = T_2 + 0.5 \text{ Hr.} = 2.5 \text{ Hr. Answer}$$

Question 100

P's work is twice as much as Q. If both work together they finish a job in 12 days. If P works alone the number of days needed to complete that work?

A 16

B 18

C 19

D 20

Answer: B

Explanation:

Solution:

A/q

Let Q's efficiency be $x\%$.

Then P's efficiency is $2x\%$

Total efficiency (if they work together) = $(x + 2x)\% = 3x\%$

With $3x\%$ efficiency, time taken = 12 days.

So, with $x\%$ efficiency time taken = $12 \times 3 = 36 \text{ days.....}$ (Efficiency is indirectly proportional to Time taken)

So time taken if P works alone @ $2x\%$ is = $36/2 \text{ days}$

= 18 days. Answer

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Question 101

A, B, C completed a work and received ₹.1800 as their wages. A worked for 6 days. B for 4 days and C for 9 days. If their wages are in the ratio of 5:6:4 the amount received by B (in rupees) is

A 600

B 720

C 480

D 500

Answer: C

Explanation:

Question 102

The length and the breadth of a rectangular field are in the ratio 3 : 2. The cost of fencing the field at the rate of ₹ 50 per meter is ₹ 25,000/-. The area of the field in square meters is

A 12,000

- B 12,225
- C 12,250
- D 15,000

Answer: D

Explanation:

Question 103

The circumference of a circle is 352 meters. Taking the approximate value of π to be $\frac{22}{7}$, the area of the circle in square meters is

- A 9546
- B 9646
- C 9756
- D 9856

Answer: D

Explanation:

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Question 104

The radius of the base of a cylinder is $\frac{1}{3}$ of the radius of the base of a cone. If their heights are equal, then the ratios of their volumes is

- A 1:3
- B 2:3
- C 2:5
- D 3:2

Answer: A

Explanation:

Question 105

Three metal cubes of sides 6, 8 and 10 cms are melted and formed into a single cube. If there is no loss of metal in the process what is the side of the big cube (in cms) ?

- A 12
- B 14
- C 15
- D 16

Answer: A

Explanation:

Question 106

A semi-circle is constructed on each side of a square of side 2m. Then the whole area of the figure (in sq.m) is

A $10^{\frac{2}{7}}$

B $10^{\frac{3}{7}}$

C $12^{\frac{2}{7}}$

D $20^{\frac{2}{7}}$

Answer: A

Explanation:

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Question 107

The difference between the circumference and radius of a circle is 111 cm. Then the area of the circle (in cm^2)?

A 693

B 1386

C 2079

D 2772

Answer: B

Explanation:

Question 108

Two poles of height 15 meters and 30 meters are erected in a playground at a distance of 36 meters between them. The distance between their tops (in meters) is

A 36

B 38

C 39

D 40

Answer: C

Explanation:

Question 109

The number equal to $(254)_5$ in ternary system is

A $(2221)_3$

B $(2121)_3$

C $(2112)_3$

D $(2211)_3$

Answer: E

Explanation:

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Question 110

The sum $(1010)_2 + (102)_3$

- A $(10011)_2$
- B $(10101)_2$
- C $(10001)_2$
- D $(10110)_2$

Answer: B

Explanation:

Question 111

For any two statements p, q, the contrapositive of $p \Rightarrow q$ is

- A $q \Rightarrow p$
- B $\sim p \Rightarrow \sim q$
- C $\sim q \Rightarrow \sim p$
- D $p \Rightarrow q$

Answer: C

Question 112

For any two statements p, q the converse of $\sim p \Rightarrow q$ is

- A $\sim (p \wedge q)$
- B $p \wedge q$
- C $p \vee q$
- D $\sim (p \vee q)$

Answer: A

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Question 113

If a N denotes the set $\{ax : x \in N\}$ then $3N \cap 7N =$

- A 21 N
- B 42 N
- C 3 N
- D 7 N

Answer: A

Question 114

If $R = \{(a, b) \in R^2 : a - b + \sqrt{2}\}$ is irrational then the relation R on R^2 is:

- A Symmetric
- B Transitive
- C Reflexive
- D An equivalence relation

Answer: D

Question 115

If $f(1) = 1$ and $f(n) = 2 \sum_{k=1}^{n-1} f(k)$, $(n > 1)$ then $\sum_{k=1}^m f(k) =$

- A $(3^m) - 1$
- B (3^m)
- C $(3^m - 2)$
- D $(3^m - 1)$

Answer: D

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Question 116

The sum of the intercepts made by the line $7x + 2y = 28$ on the coordinate axes is

- A 9
- B 18
- C 14
- D $\frac{9}{14}$

Answer: B

Question 117

A line l passing through the point (1, -2) is perpendicular to the line $x + 2y + 1 = 0$. Then the point of intersection of l and $x + 2y + 1 = 0$ is:

- A (-1, 0)
- B (1, -1)
- C $(\frac{7}{5}, \frac{-6}{5})$
- D (5, -3)

Answer: C

Question 118

$\cos(A + 45^\circ) + \sin(A - 45^\circ) =$

A $\sqrt{2} \sin A$

B $\sqrt{2} \cos A$

C 0

D 1

Answer: C

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Question 119

$$\cos^2 A = \sin A \tan A \Rightarrow \cot^6 A - \cot^2 A =$$

A 1

B 2

C 3

D -1

Answer: A

Question 120

$$\frac{\cos \theta}{7} = \frac{\sin \theta}{6} \Rightarrow 7 \cos 2\theta + 6 \sin 2\theta =$$

A 6

B $\frac{25}{17}$

C 7

D $\frac{17}{25}$

Answer: C

Question 121

The angle of elevation of a tower from a point 40 meters from foot of the tower is $\cot^{-1} \left(\frac{3}{5} \right)$. Then the height (in meters) of the tower is

A $33\frac{1}{3}$

B 40

C 50

D $66\frac{2}{3}$

Answer: D

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Question 122

The quadratic polynomial in x which when divided by $x - 1$, $x + 2$, $x + 1$ leave remainders 3, 24, 9 respectively is

- A $x^2 + x + 22$
- B $x^2 + 3x + 11$
- C $4x^2 + 3x - 2$
- D $4x^2 - 3x + 2$

Answer: D

Question 123

If $x + a$ is the highest common factor of $x^2 + bx + c$ and $x^2 + dx + e$, then the value of a is

- A $\frac{c-e}{b-d}$
- B $\frac{b-d}{c-e}$
- C $\frac{c+e}{b+d}$
- D $\frac{b+d}{c+e}$

Answer: A

Question 124

If $p(x)$ is a quadratic polynomial such that $P(1) = 7$, $p(-1) = 3$, $p(0) = 4$, then $p(10) =$

- A 110
- B 120
- C 124
- D 69

Answer: C

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Question 125

The remainder when $x^4 + 4x^3 - 5x^2 - 6x + 7$ is divided by $x - 3$, is

- A 143
- B 131
- C 132
- D 133

Answer: D

Question 126

If a number x exceeds its reciprocal by $\frac{2}{3}$, that $x^2 =$

- A 4
- B 9
- C 16
- D 25

Answer: B

Question 127

A fraction becomes $\frac{5}{7}$ if 2 is added to both its numerator and denominator. However, if 17 is subtracted from both numerator and denominator the fraction becomes $\frac{8}{15}$. If 1 is added to its numerator and denominator, the fraction becomes.

- A $\frac{17}{24}$
- B $\frac{16}{25}$
- C $\frac{18}{25}$
- D $\frac{15}{22}$

Answer: A

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Question 128

The sum of first three terms of a geometric progression is $\frac{13}{12}$ and their product is -1. Then the fourth term is

- A $-\frac{16}{9}$
- B $-\frac{8}{9}$
- C $-\frac{8}{3}$
- D $-\frac{4}{3}$

Answer: A

Question 129

How many first few terms of the geometric progression 3, 6, 12, 24.... add up to 765?

- A 15
- B 16
- C 7
- D 8

Answer: D

Question 130

The coefficient of x^5 in the expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^9$ is

A $\frac{9 \times 2^{14}}{5^6}$

B $\frac{9 \times 2^{15}}{5^6}$

C $\frac{9 \times 2^{16}}{5^5}$

D $\frac{9 \times 2^{14}}{5^5}$

Answer: D

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Question 131

$$10C_0 + 10C_2 + \dots + 10C_{10} =$$

A 2^{10}

B 2^8

C 2^9

D 2^{11}

Answer: C

Question 132

$$\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \Rightarrow 3x - 4y =$$

A 3

B 4

C 5

D 6

Answer: D

Question 133

The system of equations:

$$x + y + z = 3$$

$$2x + 5y + 4z = 11$$

$$5x + 2y + z = 10$$
 has

A unique solution

B no solution

C infinitely many solutions

D two solutions

Answer: A

Question 134

$$\lim_{x \rightarrow 0} \left(\frac{e^{5x} - e^{3x}}{e^{4x} - e^x} \right) =$$

- A $\frac{3}{2}$
- B $\frac{2}{3}$
- C $\frac{3}{4}$
- D $\frac{4}{3}$

Answer: B

Question 135

$$(x + y)^{a+b} = x^a y^b \Rightarrow \frac{dy}{dx} =$$

- A $\frac{x+y}{x-y}$
- B $\frac{y}{x}$
- C $\frac{x}{y}$
- D $\frac{x-y}{x+y}$

Answer: B

Question 136

In a $\triangle ABC$, $\angle B = 55^\circ$, $\angle C = 45^\circ$ and bisector of $\angle A$ meets BC at a point D. Then $\angle CAD =$

- A 80°
- B 40°
- C 50°
- D 60°

Answer: B

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Question 137

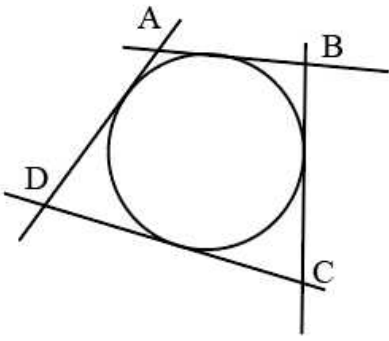
If the point A, B, C and D are Concyclic, then $\angle ABC + \angle ADC =$

- A 90°
- B 120°
- C 180°
- D 135°

Answer: C

Question 138

A circle touches all the four sides of a quadrilateral ABCD as shown below. If $AB = 3$ cm, $BC = 4$ cm and $CD = 5$ cm then $AD =$



A 4 cm

B 5 cm

C 3 cm

D 6 cm

Answer: A

Question 139

The points A (5, 2), B (6, -15), C (0, 0) are vertices of the triangle which is

A equilateral

B isosceles

C right angled

D right angled isosceles

Answer: C

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Question 140

If three vertices of a rectangle are (4, 1), (7, 4) and (13, -2) then the fourth vertex is

A (10, -5)

B (-10, -5)

C (-10, 5)

D (10, 5)

Answer: A

Question 141

The arithmetic mean of 10 observations is 48. If two more observations 65 and 75 are added to the data. then the arithmetic mean of the entire data of 12 observations is:

A $51\frac{2}{3}$

B $50\frac{2}{3}$

C $52\frac{1}{5}$

D $53\frac{3}{5}$

Answer: A

Explanation:

The mean of 10 observation is 48

Likewise the mean if the two added observation is 70($65+75/2=70$)

Using Alligation method:

Step 1: 10 2

48 70

?

Step 2: 48 70

?

5 : 1

Step 3: $5+1=22$

$1=11/3$

so as $1=11/3$, so $5=55/3$

putting those values we get ,

Step4: 48 70

?

$55/3 : 11/3$

Step 5: 48 70

$155/3$ OR $15\frac{2}{3}$

$55/3$ $11/3$

Question 142

The median of first $2n$ natural numbers is

A n

B $n + \frac{1}{2}$

C $n + 1$

D $n + 2$

Answer: B

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Question 143

For a data the median is 53.5 and the mean is 53.8. Then mode of the data is

- A 51.3
- B 52.7
- C 52.9
- D 53.4

Answer: C

Question 144

The standard deviation of first 25 natural numbers is

- A $\sqrt{54}$
- B $\sqrt{26}$
- C $\sqrt{13}$
- D $2\sqrt{13}$

Answer: D

Question 145

The variance of first n natural numbers is

- A $\frac{n^2-1}{12}$
- B $\frac{n^2-1}{6}$
- C $\frac{n^2-1}{3}$
- D $\frac{n^2-1}{24}$

Answer: A

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Question 146

If the sum of squares of the deviations of ranks of two players in n sports is $\frac{2n(n+1)}{3}$, then the coefficient of rank correlation is

- A $\frac{n-4}{n-1}$
- B $\frac{n-5}{n-1}$
- C $\frac{n-3}{n-1}$
- D $\frac{n-5}{n+1}$

Answer: B

Question 147

If A and B are two events in a random experiment with the probabilities $P(A) = \frac{3}{8}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$, then $P(A \cap B) =$ Here A is the complementary event of A

A $\frac{13}{24}$

B $\frac{11}{24}$

C $\frac{7}{24}$

D $\frac{5}{24}$

Answer: A

Question 148

If a number is selected at random out of the first 120 natural numbers, then the probability that it is a composite number is

A $\frac{89}{120}$

B $\frac{87}{120}$

C $\frac{31}{120}$

D $\frac{1}{4}$

Answer: A

Explanation:

1 is neither composite nor prime no.

Eliminating the remaining prime no from the first 120 natural numbers we get 89 numbers which are composite numbers,

Hence, $\frac{89}{120}$

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Question 149

If three vertices are selected at random from the vertices of an octagon, then the probability that no two of their three vertices are adjacent to each other is

A $\frac{2}{7}$

B $\frac{3}{7}$

C $\frac{4}{7}$

D $\frac{5}{7}$

Answer: D

Question 150

If four boys and five girls sit in a row at random, then the probability that all boys sit together is

A $\frac{1}{21}$

B $\frac{2}{21}$

C $\frac{1}{7}$

D $\frac{4}{21}$

Answer: A

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Communication Ability

Instructions

Choose the correct meaning for the word given:

Question 151

Restive

- A restful
- B restless
- C quiet
- D laid-back

Answer: B

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Question 152

Wield

- A handle
- B farm
- C extend
- D increase

Answer: A

Question 153

Whimsical

- A repulsive
- B impulsive
- C balanced
- D optimistic

Answer: B

Question 154

Vociferous

- A silent
- B friendly
- C loud
- D proud

Answer: C

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Question 155

Uprising

- A promising
- B rebellion
- C apprehension
- D approval

Answer: B

Question 156

Rampant

- A restrained
- B unsatisfactory
- C satisfactory
- D unrestrained

Answer: D

Instructions

Fill in the blank choosing the correct word:

Question 157

The wealth of India tempted many foreign invaders.

- A scanty
- B inadequate
- C fabulous
- D salubrious

Answer: C

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Question 158

The disaster was, thanks to timely action.

- A imminent
- B averted
- C certain
- D inescapable

Answer: B

Question 159

The havoc caused by the storm was

- A attractive
- B phenomenal
- C encouraging
- D animating

Answer: B

Question 160

The welfare measures by the Government benefited the entire society.

- A suppressed
- B generated
- C stalled
- D isolated

Answer: B

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Instructions

Choose the correct answer

Question 161

The package of MS Office that is used to curate documents is known as

- A Word
- B Excel
- C PowerPoint
- D Internet

Answer: A

Question 162

A barcode reader is an example for

- A** Output device
- B** Input device
- C** Processing device
- D** Storage device

Answer: B

Question 163

Algorithm consists of

- A** a set of discontinuous steps
- B** a set of sequential steps
- C** a set of imprecise steps
- D** a set of random ssteps

Answer: B

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Question 164

Which of the following is not an operating system?

- A** Microsoft
- B** Linux
- C** Windows
- D** Ios

Answer: A

Question 165

Which of the following provide bi-directional communication but in one direction at a time?

- A** Simplex
- B** Halt-duplex
- C** Full duplex
- D** Half-simplex

Answer: B

Question 166

The abbreviation 'NAV' of mutual funds stands for

- A Net Asset Value
- B Net Amount Value
- C Non Accrued Value
- D Non Admissible Value

Answer: A

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Question 167

A Security agent against a loan is known as

- A Collateral
- B Mortgage
- C Commercial paper
- D Invoice

Answer: A

Question 168

Thinking creatively, abandoning all pre conceptions is often described as thinking

- A Beyond the box
- B Beyond the sky
- C Outside the box
- D Cogently

Answer: C

Question 169

The Empirical School of Management emphasizes on the

- A Experience of managers
- B Experience of the subordinates
- C Experience of the consumers
- D Experience of the public

Answer: A

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Question 170

The wage system wherein a definite sum of money is paid for a particular of work is known as

- A The piece wage system

- B The time wage system
- C The differential piece wage system
- D The incentive wage system

Answer: B

Question 171

A: "Are you worried about Ramya"

B: "No, not at all! I know she can give as good as she gets." 'B' implies that

- A Ramya pays compliments to all
- B Ramya is very generous
- C Ramya hates every one
- D Ramya can argue or fight as strongly as one's opponent does.

Answer: D

Question 172

A: "We thought everything was settled, but now they say they are not happy."

B: "So you are back to square one again." 'B' implies that

- A the deal is partly a success
- B the deal is given up
- C the deal is back to where they started
- D the deal is settled

Answer: C

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Question 173

"Enter it in the register." Change the above sentence into passive voice.

- A Let it be entered in the register
- B It should be entered in the register
- C Find the register
- D Has it been entered in the register?

Answer: A

Question 174

"I tried to explain, but the manager cut me short." According to the speaker the manager

- A allowed him to complete what he was saying
- B questioned him on his explanation

C interrupted him and did not allow him to finish

D scolded him roughly

Answer: C

Question 175

A: "How your weekend?"

B: "It was a roller coasterride." 'B' implies that the weekend was

A unexciting

B characterized by unexpected sudden change

C strenuous

D spent in entertainment

Answer: B

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Question 176

"If the room had been brighter, I would have been able to read for a while before bed time." The speaker implies

A the room had light

B the room had light enough to read

C the room had no light at all

D the room had light but not enough to read

Answer: D

Question 177

"The manager kept this staff at a distance." The manager implies that the manager

A asked his staff to stand at a distance

B asked his staff to stay at a distance

C was not friendly with his staff

D was friendly with his staff.

Answer: C

Instructions

Fill in the blanks with the appropriate phrase/verbs/ preposition:

Question 178

"Are you hungry? Would you care something to eat?"

A for

B of

C about

D on

Answer: A

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Question 179

His salary is not enough to live

A by

B of

C for

D on

Answer: D

Question 180

The staff are strike.

A on

B by

C in

D through

Answer: A

Question 181

In a fit of emotion he his job.

A threw up

B threw over

C turned over

D tuned up

Answer: A

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Question 182

They were asked to the decision of their supervisors.

A abide in

B abide with

C abide for

D abid for

Answer: A

Question 183

If you make a mistake on the form, just

A put it off

B cross it out

C tum it off

D turn it away

Answer: B

Question 184

Engineers a new channel for the stream to follow.

A gouged

B goaded

C steered

D slithered

Answer: A

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Question 185

Do her parents about it?

A knew

B knows

C know

D known

Answer: C

Instructions

Read the following passage and answer question:

What are the economic impediments to greater financial inclusion? Perhaps the most important is the economic condition of the excluded. World over, the poor, the small and the remote are excluded. It is not just because the financial system is underdeveloped, but because they are hard to service profitability.

Nevertheless, this is not a reason to abandon hope, but to ask how we can overcome the impediments in the way of inclusion.

The excluded may live in remote areas or may belong to communities or segments of society that undertake economic activity informally—they do not maintain records or have signed contracts or documentation. They often do not own property or have regular established sources of income. As a result, a banker, especially if he is not from the local region, will have difficulty getting sufficient information to offer financial products.

Question 186

Whom does the passage exclude in the category of "the excluded"?

- A the small
- B the poor
- C the illiterate
- D the remote

Answer: C

Question 187

What is the main reason for economic exclusion?

- A the backwardness of the people
- B the difficulty in offering profitable financial services
- C the lack of money with the poor
- D the undeveloped financial resources

Answer: B

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Question 188

Which is not stated as a common feature of the excluded?

- A not having a regular source of income
- B not owning property
- C doing financial transactions informally
- D being financially illiterate

Answer: D

Question 189

Why does the banker have difficulty in getting sufficient information about his customers?

- A he does not understand their language
- B he belongs to the same region as his customers
- C he is not local to the region
- D he does not have time

Answer: C

Question 190

What does the term 'impediments' mean?

- A drawbacks
- B apparatus
- C barriers
- D implications

Answer: C

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Instructions

Read the following passage and answer questions:

Every Olympic athlete, every leader, and every human being has a little known brain part in common: a Quit Switch. Some people out of lifelong habit throw the quit switch at the first sign of frustration. Their day of phone calls gets frustrating, so they throw the switch and go for coffee with a co-worker for two hours of sympathetic negativity. Everyone has a Quit Switch. Not everyone knows it. Get to know it: notice yourself flipping the switch. You can't quit and you won't quit until you throw the switch. A human being is built like any animal to persist until a goal is reached. Somewhere along the way, though, we learn about this little switch. Soon, we start flipping the switch. We quit. If you weren't in the habit of throwing the switch too early, you would achieve virtually any goal you ever set. Whether you flip it early or late is only habit.

The switch flipping habit is misinterpreted as lack of will power, courage, drive or desire, but that is nonsense. It's a habit and like any other habit, it can be replaced with another habit. Make it your habit not to throw the Quit Switch early in any process.

Question 191

What is a "Quit Switch"?

- A an electric point
- B a sudden desire to give up
- C a portable gadget
- D a sudden feeling of happiness

Answer: B

Question 192

What are most people unaware of ?

- A that no one is infallible
- B that success comes to one and all
- C that permission is uncommon
- D that failure is inevitable

Answer: A

Question 193

When do people use the 'Quit Switch'?

- A when they are too lazy
- B when they are too busy

- C when they are frustrated
- D when they do not want to work

Answer: C

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Question 194

Why is it unnatural to 'Quit'?

- A it is a flaw
- B it is a solution
- C it is normal
- D persistence is in-built in humans

Answer: D

Question 195

Why can the 'Quit Switch' be overcome?

- A because it is a habit
- B because it is lack of will power
- C because it is lack of courage
- D because it is lack of motivation

Answer: A

Instructions

Read the following and answer questions:

From the very first day of my stay with him Gokhale made me feel completely at home. He treated me as though I were his younger brother. He acquainted himself with all my requirements and arranged to see that I got all I needed. Fortunately my wants were few and as I had cultivated the habit of self — help. I needed very little personal attendance. He was deeply impressed with my habit of tending for myself, my personal cleanliness, perseverance and regularity, and would often overwhelm me with praise.

He seemed to keep nothing private from me. He would introduce me to all the important people that called on him. This is how he introduced Dr. Ray: "This is Prof. Ray who having a monthly salary of Rs. 800. keeps just Rs.40 for himself and devotes the balance to public purposes. He is not. and does not want to get "married."

To see Gokhale at work was as much a joy as an education. He never wasted a minute. His private relations and friendships were all for public good. India's poverty and subjection were matters of constant and intense concern to him.

Question 196

According to the passage. how old is the author?

- A as old as Gokhale
- B older than Gokhale
- C younger than Gokhale
- D younger than Gokhale's brother

Answer: C

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Question 197

Why were the author's requirements few?

- A he was a clean man
- B he was self-reliant
- C he was a public figure
- D he was a guest at Gokhale's house

Answer: B

Question 198

How much did Dr. Ray spend every month for purpose?

- A Rs. 800
- B Rs. 40
- C Rs. 760
- D Rs. 740

Answer: A

Question 199

What does the phrase "called on" in the passage mean?

- A phoned
- B spoke to
- C called him by name
- D visited

Answer: D

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Question 200

Identify the false statement about Gokhale.

- A He never wasted his time
- B He was never concerned with India's freedom
- C He was deeply concerned with India's poverty
- D He worked for public good

Answer: B

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